

Visvesvaraya Technological University
Belgaum, Karnataka- 590014



Third Year
A Mini-Project Report

On

**“ Explainable Multimodal Brain – Heart Stress Detection and
AI-Driven Intervention System using EEG and ECG Signals ”**

Submitted in the partial fulfilment of the requirements for the award of the Degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING
(DATA SCIENCE)

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2025–2026

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2025–2026

Certificate

This is to certify that the Mini Project Work entitled **“Explainable Multimodal Brain – Heart Stress Detection and AI-Driven Intervention System using EEG and ECG Signals”** is a bonafide work carried out by **Ashlesh (1DS23CD012), Sambhram S Naregal (1DS23CD045), Tushar D Kudachimath (1DS23CD058) and Vishal Krishna Bhat K (1DS23CD063)** , in partial fulfilment for the VI semester of Bachelor of Engineering in CSE (Data Science) of the Visvesvaraya Technological University, Belgaum during the year 2025–2026. The Project report has been approved as it satisfies the academic requirements prescribed for the Bachelor of Engineering degree.

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Name of the Examiners

Signature with Date

1.

2.

ACKNOWLEDGEMENT

First and foremost, we thank the Almighty for being our light and for showering his gracious blessings throughout this project.

We express our gratitude to our beloved Principal, Dr. B.G. Prasad, for providing us with all facilities.

With a grateful heart, our sincere thanks to our Head of the Department, Dr. Harish H. Kenchannavar, Department of Computer Science Engineering (Data Science), for the motivation and support to complete the project work.

We extend our sincere thanks to our project guide, Prof./Dr. Dr. Aarathi S, Associate Professor, Dept. of CSE (Data Science), DSCE, and to our project coordinator Prof./Dr. Dr. Aarathi S, for the valuable guidance and suggestions in all aspects that aided us in ameliorating our skills.

We are thankful to all the Teaching and Non-Teaching Staff of the Department of Computer Science Engineering (Data Science) and to all those who have directly and indirectly extended their help to us in completing this project work successfully.

We extend our sincere thanks to our family members and our beloved friends, who had been strongly supporting us in all our endeavors.

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Chapter 1

Template Usage Guide

This chapter contains sample examples for commonly used LaTeX components in this template.

1.1 Section Example

This is a normal section example.

1.1.1 Subsection Example

This is a subsection example.

1.2 Bullet List Example

- Point One
- Point Two
- Point Three

1.3 Numbered List Example

1. First Point
2. Second Point
3. Third Point

1.4 Table Example

Table 1.1: Sample Table

Column 1	Column 2
Data A	Data B
Data C	Data D

1.5 Image Example



Figure 1.1: Sample Image

1.6 Citation Example

Add references inside references.bib

Example:

```
@article{r1,  
author = {John Smith},  
title = {Artificial Intelligence in Clinical Systems},  
journal = {Journal of Medical Computing},  
year = {2024}  
}
```

Use citations like:

`\cite{r1}`

Example:

Artificial Intelligence improves healthcare systems [1].

1.7 Code Example

```
print("Hello World")
```

1.8 Equation Example

$$E = mc^2$$

1.9 Important Notes

- Put all images inside the images/ folder
- Compile multiple times if TOC or references do not update
- Add references inside references.bib

Chapter 2

Introduction

Write the complete introduction of the project here.

Chapter 3

Literature Survey

Discuss existing systems and related research.

Chapter 4

Research Gap Identified

Explain the limitations in existing systems.

Chapter 5

Architectures

5.1 High Level Architecture

Insert high level architecture diagram here.

5.2 Detailed Architecture

Insert detailed architecture diagram here.

5.3 UML Diagrams

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- Activity Diagram

Chapter 6

Project Objectives

- Objective 1
- Objective 2
- Objective 3
- Objective 4

Chapter 7

Timeline using Gantt

Insert Gantt chart here.

Chapter 8

Hardware and Software Requirements

8.1 Hardware Requirements

- Processor
- RAM
- Storage

8.2 Software Requirements

- Operating System
- Programming Language
- Frameworks
- Database

Chapter 9

Sample Code

Example code here

Chapter 10

Project Demo Screenshots

Insert screenshots here.

Chapter 11

AI Tools Application

Explain AI tools used in the project.

Chapter 12

SGDs Met by Project with Proper Justification

Mention SDGs achieved by the project.

Chapter 13

Purpose of the Project

13.1 Purpose of the Proposed System

The proposed system aims to develop an explainable multimodal framework for stress detection using EEG and ECG physiological signals. By combining brain and heart activity analysis, the system attempts to improve both the accuracy and interpretability of stress prediction.

In addition to stress classification, the framework also focuses on generating personalized stress management recommendations using Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs).

13.2 Targeted Audience

The project is mainly intended for research and clinical-support environments where EEG and ECG acquisition systems are available.

The proposed framework can be useful for:

- Healthcare professionals and mental health researchers
- Biomedical signal processing researchers
- AI and Explainable AI researchers
- Hospitals and clinical research laboratories
- Wearable healthcare technology developers

The system can also serve as a foundation for future wearable stress monitoring technologies.

13.3 Expected Outcome of the Project

The expected outcome of the project is the development of a multimodal stress analysis framework capable of:

- Performing five-level stress classification
- Providing Explainable AI-based interpretation
- Identifying important physiological stress indicators
- Generating evidence-based stress management recommendations

The framework contributes towards research in:

- Healthcare Artificial Intelligence
- Explainable AI (XAI)
- Biomedical signal processing
- Mental health monitoring systems

13.4 Research and Future Scope

The proposed project contributes towards ongoing research in the fields of Healthcare Artificial Intelligence, Explainable AI (XAI), and Biomedical Signal Processing. By combining EEG and ECG-based multimodal analysis with Transformer architectures and explainability techniques, the framework attempts to improve both stress prediction performance and interpretability.

The integration of Retrieval-Augmented Generation (RAG) and LLM-based intervention planning further extends the system beyond conventional stress classification by providing evidence-based personalized recommendations.

The proposed framework can be further extended through:

- Integration with wearable healthcare devices
- Inclusion of additional physiological modalities
- Larger real-world clinical validation
- Real-time stress monitoring applications

The work also has scope for future research publication and potential development into advanced healthcare support systems.

Chapter 14

Research Paper Draft

14.1 Research Paper Title

An Explainable Multimodal Transformer Framework for EEG–ECG-Based Stress Detection with RAG-Enabled Personalized Intervention

14.2 Abstract

Stress is a prevalent issue worldwide which impacts one's psychological and physical health. Recent developments have been made in the detection of stress through wearables which measure physiological signals such as Electroencephalography (EEG), Electrocardiogram (ECG), Electrodermal Activity (EDA), Galvanic Skin Response (GSR), and Photoplethysmography (PPG). Machine learning models utilized for prediction include Logistic Regression, KNN, Random Forest, Support Vector Machines (SVM), XGBoost, and Neural Networks. In order to provide an explanation and make the predictions more understandable, Explainable Artificial Intelligence (XAI) methods like SHAP, LIME, and attention mechanisms are utilized.

14.3 Key Areas Covered

The research paper includes the following major areas:

- Literature survey on stress detection and Explainable AI
- Multimodal EEG and ECG signal analysis
- Transformer-based stress classification
- Explainability techniques for physiological interpretation
- Retrieval-Augmented Generation (RAG) integration

- Personalized stress management planning
- Result analysis and system discussion

14.4 Research Contribution

The proposed work contributes towards research in:

- Healthcare Artificial Intelligence
- Explainable AI (XAI)
- Biomedical Signal Processing
- Mental Health Monitoring Systems

The work also demonstrates the integration of multimodal physiological analysis with explainable and evidence-grounded AI systems for stress assessment and intervention planning.

References and Bibliography

- [1] John Smith. Artificial intelligence in clinical systems. *Journal of Medical Computing*, 2024.